

# Code walkthrough — 15 minutes

Updated 2026-09-10 for the government asset layer — the inflation-linked engine, the agency and guaranteed wrappers, and a corrected module map.

Updated 2026-09-03 for the government and municipal classes.

Updated 2026-08-31 (Round 2b: floating, fixed-then-floating and stepped bonds; instrument-type dispatch; the delivery-quality pass).

Speaking aid for the session with Mario and the engineering team, 2026-08-25. Supersedes the 2026-08-17 version, which covered only the plain-bond chain. Everything claimed below is backed by the automatic suite: 223 checks, about 19 seconds.

**Opening line:** "The code is organised so that the mathematics lives in one place and each bond type is a short file that just names its inputs. Everything from the August sample still produces identical numbers — and there are now three more bond types and two ways to call it from outside Python. Here is the tour."

## Where each legacy engine went — the reconstruction map

Read this first if the question is "is this the same code that produced last month's numbers?" The answer is yes, and each line is checked by a test rather than by inspection.

| Legacy file                | New core location                                    | Old path | Asset wrapper                 | Endpoint type      |
|----------------------------|------------------------------------------------------|----------|-------------------------------|--------------------|
| pricing/bond_price.py      | core/pricing/analytical.py + cashflows + discounting | shim     | assets/corporate/vanilla.py   | vanilla            |
| pricing/lattice.py         | core/pricing/tree.py                                 | shim     | callable / puttable / sinking | those three        |
| pricing/frn.py             | core/pricing/floating.py                             | shim     | assets/corporate/floating.py  | floating           |
| pricing/hybrid.py          | core/pricing/hybrid.py                               | shim     | assets/corporate/hybrid.py    | fixed_to_floating  |
| pricing/coupon_schedule.py | core/pricing/coupon_schedule.py                      | shim     | assets/corporate/stepped.py   | stepped            |
| pricing/ilb.py             | core/pricing/inflation.py                            | shim     | assets/government/linker.py   | none — by decision |

⚠ The last row is the one to pause on. The inflation-linked engine moved on 2026-09-10 and has a wrapper like every other product, but **deliberately no endpoint type**: the spreadsheet has no cells for a real coupon, an index ratio or an inflation assumption, and a type becomes reachable from a worksheet only once Mario has chosen that layout. The engine and contract support seven types; the Excel bridge can construct five. That gap is a decision, not a backlog, and a test pins it so it cannot drift open.

Four claims, each pinned:

1. the numerical body is **byte-identical** below the docstring (hybrid's two import lines are the one exception, and both targets are asserted to be *the same objects*);
2. every shim re-exports **the same object** — `a is b`, not an equal reimplementaion;
3. every endpoint result equals the direct wrapper call with `==`, per instrument type;
4. the production output files are compared **by hash**, before and after every commit.

And the path a bond actually takes, end to end:

|                         |                                                                          |
|-------------------------|--------------------------------------------------------------------------|
| workbook coupon family  | Coupon_Formula2 on the Corporate Bonds tab                               |
| -> router / overrides   | dataio/coupon_types.py, dataio/term_overrides.py                         |
| -> core engine          | core/pricing/{analytical,tree,floating,hybrid,coupon_schedule,inflation} |
| -> asset wrapper        | assets/{corporate,government}/*.py units + named inputs, no arithmetic   |
| -> endpoint type        | endpoints/ one request in, one response out                              |
| -> driver / JSON output | scripts/*.py                                                             |
| -> evidence             | tests + hashed CSVs + the disposition sidecars                           |

**Government and municipal bonds (added 2026-09-03) reuse all of the above and add no engine.** They enter through the same loader ( `dataio/phase2.py` , which already served the agency, guaranteed and index-linked classes), route to the analytical pricer or, for the one

callable US Treasury, to the same option tree, and come out of a separate driver, `scripts/sovereign_risk.py`. The driver is separate on purpose: `phase2_risk.py` writes one of the five files we compare by hash after every commit, and adding 154 rows to it would have spent that guarantee for nothing. The shared loader therefore takes an explicit class list whose default is the original four, so that file is untouched. The only genuinely new code is eight currency-to-curve-file registrations and a price-quotation resolver.

### What changed numerically, in one place

|                                                                           |                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| moving the three engines                                                  | nothing at all – five driver CSVs byte-identical                                                                                                                                                                                                                                                                                                                                      |
| the exception family, the ACT/364 conversion, the private-helper refactor | nothing at all – five driver CSVs byte-identical                                                                                                                                                                                                                                                                                                                                      |
| the GBP par-yield units correction                                        | two GBP securities, and the headline counts                                                                                                                                                                                                                                                                                                                                           |
| the callable routing fix                                                  | no priced number; one bond went from invisible to named                                                                                                                                                                                                                                                                                                                               |
| the defaulted-security rule                                               | no priced number; one bond (8.78M par) went from invisible to a named recovery row. 565 -> 566 @3-31                                                                                                                                                                                                                                                                                  |
| the FRN current-coupon freeze                                             | THE one intentional change: six floating notes' rate sensitivity. No price, no spread, no other column, and no other file. Table in <code>docs/frn_current_coupon_freeze_2026-08-31.md</code>                                                                                                                                                                                         |
| the provenance labels                                                     | new COLUMNS only; every pre-existing value unchanged except one FRN flag string                                                                                                                                                                                                                                                                                                       |
| the government/municipal classes (2026-09-03)                             | nothing at all in any existing file – two NEW outputs, and all five driver CSVs plus all four disposition sidecars regenerate byte-identical                                                                                                                                                                                                                                          |
| the curve-failure message fix                                             | no number anywhere; the message had been embedding a filesystem path, so the same failure read differently on Windows and on Linux                                                                                                                                                                                                                                                    |
| the government asset layer (2026-09-10)                                   | nothing at all, anywhere. The inflation-linked engine moved, three wrapper modules were added, the driver switched to the new import path and gained the representability guard -- and all 22 files in outputs/ regenerate byte-identical after re-running every driver. The guard refused nothing: all five agency call schedules reach an exercise node, so it is latent, not live. |

Everything above is proven by hashes rather than by inspection: `docs/release_facts_2026-09-10.md` carries the sha256 of all seven production CSVs and all six disposition sidecars.

### The tour, in order

#### 1. `src/pricer/__init__.py` — the map (30 seconds)

Three layers, and the split is the point:

|                         |                                                   |                    |
|-------------------------|---------------------------------------------------|--------------------|
| <code>core/</code>      | the mathematics, shared by every bond type        | (~80% of the code) |
| <code>assets/</code>    | one short file per bond type, no arithmetic       | (~20%)             |
| <code>endpoints/</code> | the outside world: one request in, one result out |                    |

Say: "If you only remember one thing — a new bond type is a new file in `assets/`, not a new engine."

#### 2. `assets/corporate/` and `assets/government/` — one file per bond type (3 min)

Open `vanilla.py`, then `callable.py`, then `floating.py`. They look the same on purpose: one simple function per output, the same names the legacy Monthly sheet uses. There are now **seven** of these files — vanilla, stepped, floating, hybrid, callable, puttable, sinking.

|                               |                          |                       |                   |                        |                       |                         |
|-------------------------------|--------------------------|-----------------------|-------------------|------------------------|-----------------------|-------------------------|
| <code>calculated_price</code> | <code>implied_oas</code> | <code>duration</code> | <code>dv01</code> | <code>convexity</code> | <code>widening</code> | <code>tightening</code> |
|-------------------------------|--------------------------|-----------------------|-------------------|------------------------|-----------------------|-------------------------|

Then point at what differs, which is only ever the bond's own terms: `callable.py` takes a **call schedule**, `puttable.py` a **put schedule**, `sinking.py` a **redemption schedule**, `hybrid.py` a **switch date and a post-switch margin**, `stepped.py` a **coupon table**. None of those files does any arithmetic; they name their inputs, convert units and pass the terms down.

`floating.py` is the one worth pausing on, because its input list differs in a way that explains the product: **there is no coupon**. A floating coupon is not known in advance, so the fixed rate is replaced by a quoted margin plus (optionally) the one coupon already fixed at the last reset. That single difference is the whole instrument.

Open one function to show the numbered `Inputs` block with units in capitals — this layer speaks the legacy sheet's units (coupon in PERCENT, prices per 100, spreads in BASIS POINTS); the engines underneath work in decimals.

**For the engineers:** every one of these is a pure function — no state, no globals, one bond per call. A portfolio is embarrassingly parallel with no design change.

### 3. `core/pricing/tree.py` — one engine, three products (4 min)

This is the heart of the new work. A callable, a puttable and a sinking-fund bond are the same calculation with a different right attached:

|            |                                                           |                                                |
|------------|-----------------------------------------------------------|------------------------------------------------|
| call right | the issuer caps the value at the call price               | <code>min(value, price)</code>                 |
| put right  | the holder floors it at the put price                     | <code>max(value, price)</code>                 |
| sinking    | the issuer retires a fraction <code>f</code> at the price | <code>(1-f)·value + f·min(value, price)</code> |

Worth saying out loud: **at `f = 1` the sinking rule is the call rule** — and the code produces the identical value to the last bit, which is one of the tests.

Why one engine and not three: fourteen rows in the workbook carry two rights at once ( `CALL/SINK` , `CALL/PUT` ). Three separate engines could not price those without copying each other.

If asked how the tree is built: it is calibrated so that it reprices the input curve's own discount factors exactly — arbitrage-free by construction, not by assumption.

### 4. `core/` — the rest, bottom-up (3 min)

- `utils/dates.py` — the calendar. One year is 364 days, one half-year 182. That is the legacy system's own convention, kept deliberately; changing it would break every validated number.
- `pricing/cashflows.py` — the cash-flow table and **the one** accrued-interest formula that every engine shares.
- `pricing/discounting.py` — the corrected discounting that reprices a curve's own par bonds to exactly 100.
- `pricing/analytical.py` — the plain-bond price: dates → cash flows → discount → sum. About 25 lines of orchestration.
- `risk/sensitivities.py` — duration, DV01, convexity as pure arithmetic on three prices. Point it at *any* pricing function; the tree uses the same one.
- `market/curves.py` — `resolve_curve(currency, date, frequency)` : the one place a bond gets its own-currency curve, and the one place a missing curve is refused rather than substituted.

### 5. `endpoints/` + `integrations/excel_vba/` — calling it from outside (3 min)

Excel cells → VBA bridge → `request.json` → `analyze_vanilla_payload()` → `response.json` → cells

`endpoints/main.py` is one function: a dictionary in, a dictionary out, and it never raises — failures come back as a status and a reason. `contracts.py` is the request/response shape, standard library only. `pricing.py` arranges the calls. **No formula anywhere in this layer.**

Two things to demonstrate if there is time: - `integrations/excel_vba/examples/` — a real request and the real response it produced; - `Run-BridgeTests.ps1` — 23 checks driving actual Excel, including a live call into Python.

**The line that matters for the cloud move:** the spreadsheet knows one setting, a command to run. Point it at a packaged executable or an HTTP service and nothing else changes.

### 6. Proof, then close (2 min)

- `src/pricing/*.py` — the old module paths are now thin shims. Every existing script, driver and test still works unchanged.
- Migration safety: production output files are compared **by cryptographic hash** before and after every change. Identical, three times over.

- `tests/test_pricer_tree_structure.py` — the interesting ones: the new wrapper equals the production calculation with `==`; a bond with no rights prices exactly as a plain bond; callable  $\leq$  plain  $\leq$  puttable.

## Numbers to have ready

| Claim                                                   | Number                                                                                                                           |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Automatic checks                                        | <b>424</b> in ~45 s (390 before this round)                                                                                      |
| Asset classes complete                                  | <b>six of the cash-bond classes — 1,040 of the book's 2,366 lines</b>                                                            |
| Government + Municipal                                  | <b>154</b> securities · <b>147</b> priced at 3-31, <b>150</b> at 6-10, <b>7</b> and <b>4</b> named                               |
| Currencies priced                                       | <b>14</b> curve files mapped, up from 6                                                                                          |
| Behaviour change from the restructuring                 | <b>zero</b> — production CSVs hash-identical                                                                                     |
| Excel bridge                                            | <b>57/57</b> fixture mode, <b>61/61</b> against a live Python engine, on real Excel                                              |
| Cross-platform                                          | response file <b>byte-identical</b> on Windows and Linux                                                                         |
| Volatility, call-active bond                            | ~10 cents of price, or ~1 bp of spread, per vol point                                                                            |
| Bond types — <b>three different numbers</b>             | <b>7</b> the engine and request format cover · <b>5</b> the spreadsheet can build · <b>5</b> verified by a real-Excel round trip |
| Corporate output                                        | <b>566</b> @3-31 (555 with a model spread + 11 at the custodian mark) · <b>561</b> @6-10                                         |
| Bonds added by the GBP units fix                        | <b>2</b> — one blocked, one silently skipped                                                                                     |
| Bonds added by the defaulted-rule fix                   | <b>1</b> — 8.78M par, matched neither of two recovery paths                                                                      |
| Exercise terms confirmed against Bloomberg              | <b>0 of 9</b> — every schedule is labelled <code>provisional</code> in the output                                                |
| Securities accounted for in the disposition files       | <b>732</b> corporate + <b>154</b> sovereign, at each of the two dates                                                            |
| Government bonds priced on their own government's curve | median spread <b>0.0 bp</b> (Japan) and <b>4.4 bp</b> (UK) — the anchor check                                                    |

## Likely questions — answers ready

- "Why 364-day years?" — the legacy engine's own convention. We keep it so every number reconciles. Real day-count labels are carried as data and can drive a future layer.
- "Why is a callable bond's duration shorter?" — the issuer's right to repay caps how much the price can rise when rates fall, so the bond moves less.
- "Can we price a bond that is both callable and sinkable?" — the engine can hold both sets of terms. Where the two land on the same date we currently refuse, because the order of the two rights changes the answer and no order is tested yet. That is a deliberate boundary, not a gap in the mathematics.
- "Why refuse instead of assuming?" — every refusal in the code is a case where a silent assumption would produce a plausible wrong number: a missing curve, a spreadsheet date sent as a number, terms that contradict each other.
- "How would this scale to a whole portfolio in the cloud?" — pure functions, one bond per call, deterministic across platforms, and one entry point that any transport can wrap.
- "What is not built yet?" — mortgages wait on the data pull; convertibles need an equity model we do not have; and the demonstration spreadsheet still offers plain bonds only, which is a layout decision rather than a technical one.
- "How do you know nothing is silently missing?" — because the count is now mechanical. Every bond in the portfolio must leave the calculation with either a number or a named reason, and that is checked as a **set of identifiers**, not as a total. A total balances even when the wrong bond is in the wrong place, which is exactly how two omissions survived. The check runs on every production run and stops it if anything is unaccounted for.
- "You said the pound curve was broken. What changed?" — we were reading that file in the wrong units. Most of the market-data files store rates as decimals; the pound file stores percentages, and we multiplied it by a hundred. The bootstrapper was right to refuse the

result; we were wrong to read its complaint as a fact about the data. Both pound bonds now price, and one of them had been missing from the output entirely.